Magnetotransport of Weyl semimetals due to the chiral anomaly

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We study the electric field and temperature gradient driven magnetoconductivity of a Weyl semimetal system. To analyze the responses, we utilize the kinetic equation with semiclassical equations of motion modified by the Berry curvature and orbital magnetization of the wave packet. Apart from the known positive quadratic magnetoconductivity, we show that due to the chiral anomaly, the magnetoconductivity can become a nonanalytic function of the magnetic field, proportional to the $\frac{3}{2}$ power of the magnetic field at finite temperatures. We also show that time-reversal symmetry breaking tilt of the Dirac cones results in linear magnetoconductivity. This is due to the one-dimensional chiral anomaly for which the tilt is responsible.

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I. INTRODUCTION

Three-dimensional Dirac and Weyl semimetals are materials whose band structure has linearly touching conduction and valence bands [1–4], the Dirac cones. A Dirac semimetal is degenerate in the electron's right and left chiralities, while the Weyl semimetal has the two chiralities split in energy or momentum. Inversion or time-reversal symmetries must be broken to obtain the splitting of chiralities in Dirac semimetals.

Theoretically, linear band touching introduces nontrivial Berry [5] curvature to the description of the fermion dynamics. The Berry curvature in this case is an effective magnetic field in $\mathbf{k}$ space which is created by a magnetic monopole located at the band touching point. For a review on the effects of Berry curvature on electronic properties, see Ref. [6].

Weyl semimetals with broken time-reversal symmetry are characterized by the anomalous Hall effect [4,7]. Due to splitting of the Dirac cones, there are chiral edge states on the physical boundaries of the system [2,4]. Apart from that there is the so-called chiral anomaly of the Dirac fermions—nonconservation of particles with a given chirality in the presence of magnetic and electric fields [8–10].

The chiral anomaly results in novel magnetotransport properties which serve as distinctive features of Weyl semimetals. In Refs. [11–13] it was shown that the anomaly contributes to longitudinal quadratic magnetoconductivity. In contrast to the Lorentz force driven negative magnetoconductivity, the one due to the anomaly was shown to be positive in sign when the fields are collinear. This novel magnetic field dependent contribution to the current flows in the direction of the applied magnetic field. Signatures of positive magnetoconductivity were experimentally observed in a number of materials believed to be Weyl semimetals. For example, these are Cd_3As_2 [14–16], Na_3Bi [17], $ZrTe_5$ [18], and TaAs [19,20].

In this paper we extend the study of the nature of the chiral anomaly induced magnetoconductivity in Weyl semimetals. Just as in Refs. [11,12], we utilize the kinetic equation to describe the transport properties of the system. The kinetic equation for the wave packet is supplemented with the equations of motion modified by the Berry curvature and orbital magnetization. We explicitly focus on the intervalley scattering processes in the collision integral, and show how these processes stabilize the steady state in the case of the chiral anomaly. For the sake of generality, we study electric currents driven by an electric field and temperature gradient. In the case of an electric field, at small magnetic fields and small temperatures, we reproduce the result of Refs. [11,12] for magnetoconductivity. However, at nonzero temperatures and larger magnetic fields, we show that the magnetoconductivity can become a nonanalytic function of the magnetic field proportional to $B^{3/2}$. The nonanalyticity is due to the singularity of the Berry curvature at small momenta, and we note that this contribution does not come from the Fermi surface. Similar nonzero temperature, nonanalytic magnetic field dependent magnetoconductivity was found in graphene [21] and in up-doped Weyl and Dirac semimetals [22].

We also compute the magnetic field dependence of electric currents driven by the temperature gradient. In this case the magnetoconductivity has quantitatively similar behavior as in the case of electric field driven currents.

We further include a time-reversal symmetry breaking tilt of the Dirac cones. We show that the tilt results in signatures of the one-dimensional chiral anomaly. Due to this there are two magnetic field dependent contributions to the electric current that are different in nature. The first contribution to the current flows in the direction of the tilt, while the second flows in the direction of the magnetic field. Both contributions are linear in the magnetic field. The tilts of the Dirac cones have recently received theoretical attention [23–25]. Various transport properties of Weyl semimetals due to tilt were studied in Refs. [26,27]. Systems with tilted Dirac cones were recently realized in Refs. [28,29]. The present paper proposes linear magnetoconductivity as a distinct feature of time-reversal symmetry broken (e.g., ferromagnetic) Weyl semimetals, which are awaiting their experimental discovery.

II. KINETIC EQUATION FOR WEYL SEMIMETALS

We consider a model of three-dimensional Weyl fermions, which consists of two valleys with corresponding $s = \pm$ chiralities. The Hamiltonian for the $s = \pm$ valley is

$$H_s = C_s k_z^{(s)} + sv\left[\sigma_x k_x + \sigma_y k_y + \sigma_z k_z^{(s)}\right] - \mu, \qquad (1)$$

where σ_i are $i = x,y,z$ Pauli matrices corresponding to a generalized spin degree of freedom. The valleys $s = \pm$ are split in momentum as $k_z^{(s)} = k_z + sQ$, where Q is the splitting. However, as further calculations show, it is safe to ignore Q in the derivations by shifting the center of the coordinates to $k_z^{(s)} = 0$ when studying the corresponding valley. Diagonalization of the Hamiltonian gives the spectrum of the fermions, $\varepsilon_{\mathbf{k}\eta}^{(s)} = C_s k_z + \eta v k - \mu$, and the band velocity is obtained to be $\tilde{\mathbf{v}}^{(s)} = C_s \mathbf{e}_z + \eta v \frac{\mathbf{k}}{k}$. Here, μ is a Fermi energy, v is a velocity, and $\eta = \pm$ denote the conduction and valence bands, correspondingly. C_s is a parameter denoting the tilt of the Dirac cones. We choose it to be antisymmetric in the valleys, $C_+ = -C_-$. We assume that $v \gg |C|$, such that the Dirac cones are elliptical due to tilt. In the following, we assume $\mu > 0$, such that all excitations occur in the conduction, $\eta = +$, band. We focus only on the $\eta = +$ band in the following and omit the η symbol altogether until mentioned.

In order to study the magnetotransport properties of the system, we utilize the kinetic equation for the distribution function $n_{\mathbf{k}}^{(s)}$ of wave packets constructed out of conduction band fermions,

$$\frac{\partial n_{\mathbf{k}}^{(s)}}{\partial t} + \dot{\mathbf{k}}^{(s)} \frac{\partial n_{\mathbf{k}}^{(s)}}{\partial \mathbf{k}} + \dot{\mathbf{r}}^{(s)} \frac{\partial n_{\mathbf{k}}^{(s)}}{\partial \mathbf{r}} = I_{\text{coll}}\big[n_{\mathbf{k}}^{(s)}\big], \tag{2}$$

where I_{coll} is the collision integral. Semiclassical equations of motion of wave packets in the presence of electric $\mathbf{E}$ and magnetic $\mathbf{B}$ fields are, for example (see Refs. [6,30] for a review),

$$\begin{aligned} \dot{\mathbf{r}}^{(s)} &= \frac{\partial \epsilon_{\mathbf{k}}^{(s)}}{\partial \mathbf{k}} + \dot{\mathbf{k}}^{(s)} \times \mathbf{\Omega}_{\mathbf{k}\eta}^{(s)}, \\ \dot{\mathbf{k}}^{(s)} &= e\mathbf{E} + \frac{e}{c}\dot{\mathbf{r}}^{(s)} \times \mathbf{B}, \end{aligned} \tag{3}$$

where $\mathbf{\Omega}_{\mathbf{k}}^{(s)}$ is the Berry curvature and where the energy of the wave packet $\epsilon_{\mathbf{k}}^{(s)} = \varepsilon_{\mathbf{k}}^{(s)} - \mathbf{m}_{\mathbf{k}}^{(s)}\mathbf{B}$ is updated by the orbital magnetization $\mathbf{m}_{\mathbf{k}}^{(s)}$ of the wave packet. Nontrivial Berry curvature and magnetization are consequences of a band degeneracy in the Brillouin zone. For example, in a Weyl semimetal there is a linear touching of the conduction and valence band at the Weyl points. In this case, one can show that the Berry curvature is a $\mathbf{k}$-space analog of the magnetic field created by a magnetic monopole located at these points, $\mathbf{\Omega}_{\mathbf{k}}^{(s)} = -s\frac{\mathbf{k}}{2k^3}$. The orbital magnetization is derived to be $\mathbf{m}_{\mathbf{k}}^{(s)} = -se\frac{v\mathbf{k}}{2ck^2}$. The Berry curvature and magnetization do not become affected by the tilt as the tilt enters the Hamiltonian with an identity matrix. One can spot an identity $\mathbf{m}_{\mathbf{k}}^{(s)} = \frac{e}{c}vk\mathbf{\Omega}_{\mathbf{k}}^{(s)}$. After straightforward transformations in a system of equations on $\dot{\mathbf{r}}$ and $\dot{\mathbf{k}}$, one gets

$$\begin{aligned} \dot{\mathbf{r}}^{(s)} &= \frac{1}{\Delta_{\mathbf{k}}^{(s)}}\Big[\mathbf{v}^{(s)} + e\mathbf{E} \times \mathbf{\Omega}_{\mathbf{k}}^{(s)} + \frac{e}{c}\big(\mathbf{\Omega}_{\mathbf{k}}^{(s)}\mathbf{v}^{(s)}\big)\mathbf{B}\Big], \\ \dot{\mathbf{k}}^{(s)} &= \frac{1}{\Delta_{\mathbf{k}}^{(s)}}\Big[e\mathbf{E} + \frac{e}{c}\mathbf{v}^{(s)} \times \mathbf{B} + \frac{e^2}{c}(\mathbf{E}\mathbf{B})\mathbf{\Omega}_{\mathbf{k}}^{(s)}\Big], \end{aligned} \tag{4}$$

where we defined $\Delta_{\mathbf{k}}^{(s)} = 1 + \frac{e}{c}(\mathbf{B}\mathbf{\Omega}_{\mathbf{k}}^{(s)})$, and the velocity is $\mathbf{v}^{(s)} = \frac{\partial \epsilon_{\mathbf{k}}^{(s)}}{\partial \mathbf{k}}$, calculated to be $\mathbf{v}^{(s)} = v\frac{\mathbf{k}}{k}[1 + \frac{2e}{c}(\mathbf{B}\mathbf{\Omega}_{\mathbf{k}}^{(s)})] + s\frac{ev}{2ck^2}\mathbf{B} + sC\mathbf{e}_z$. We note that the semiclassical equations of motion are obtained from a quantum kinetic equation by expanding the latter in a small parameter $\frac{\omega_c}{\mu} < 1$, where $\omega_c = v\frac{eB}{ck_F}$, with k_F being the Fermi momentum [31,32]. When $k < \sqrt{\frac{eB}{c}}$, a Landau quantization of the energy levels must be made. This is a completely different problem and is out of the scope of the present paper.

The collision integral is assumed to only contain electron-impurity scattering. The system of our study consists of two valleys, hence in general there are two distinct electron-impurity scattering processes, namely, intervalley and intravalley. Both can be conveniently captured by a collision integral with a Berry curvature modified density of states [6],

$$I_{\text{coll}}\big[n_{\mathbf{k}}^{(s)}\big] = -\int_{\mathbf{k}'} \Delta_{\mathbf{k}'}^{(s')}\omega_{\mathbf{k}',\mathbf{k}}^{(ss')}\big[n_{\mathbf{k}}^{(s)} - n_{\mathbf{k}'}^{(s')}\big]\delta\big(\epsilon_{\mathbf{k}}^{(s)} - \epsilon_{\mathbf{k}'}^{(s')}\big), \tag{5}$$

where $\omega_{\mathbf{k}',\mathbf{k}}^{(ss')}$ is the probability of electron-impurity scattering between the s and s' valleys. For short range impurities we obtain the collision integral (see a comment in Ref. [33])

$$I_{\text{coll}}\big[n_{\mathbf{k}}^{(\pm)}\big] = \frac{\bar{n}^{(\pm)} - n_{\mathbf{k}}^{(\pm)}}{\tau} + \frac{\bar{n}^{(\mp)} - n_{\mathbf{k}}^{(\pm)}}{\tau_{\mathrm{V}}}, \tag{6}$$

where $\bar{n}^{(s)} = \langle \Delta_{\mathbf{k}}^{(s)} n_{\mathbf{k}}^{(s)} \rangle$, and here we defined $\langle \cdots \rangle = \frac{1}{4\pi}\int \sin(\theta)d\theta d\phi(\cdots)$, τ is the intravalley scattering time, and τ_{V} is the intervalley scattering time. We assume that the scattering time is isotropic, i.e., angle independent. It is a valid assumption for the $v \gg |C|$ case. It is convenient to rewrite the collision integral as

$$I_{\text{coll}}\big[n_{\mathbf{k}}^{(s)}\big] = \frac{1}{\tau^*}\big[\bar{n}^{(s)} - n_{\mathbf{k}}^{(s)}\big] + \Lambda^{(s)}, \tag{7}$$

where we introduced an important quantity,

$$\Lambda^{(\pm)} = \frac{1}{\tau_{\mathrm{V}}}[\bar{n}^{(\mp)} - \bar{n}^{(\pm)}], \tag{8}$$

and $\tau^* = \frac{\tau\tau_{\mathrm{V}}}{\tau + \tau_{\mathrm{V}}}$ is the total scattering time.

Let us now approximate the kinetic equation. Assume that due to the scattering lifetime, the distribution function is static, i.e., there is a steady state in the system. That allows one to drop the time derivatives in the kinetic equation. Multiply the rest of the kinetic equation for the $s = \pm$ valley by the $\Delta_{\mathbf{k}}^{(s)}$ quantity and average over the angles. After that, the $\propto\frac{1}{\tau^*}$ term in the collision integral (7) vanishes. The remaining equation reads as

$$\Lambda^{(s)} = \left\langle \Delta_{\mathbf{k}}^{(s)}\dot{\mathbf{k}}^{(s)}\frac{\partial n_{\mathbf{k}}^{(s)}}{\partial \mathbf{k}} \right\rangle + \left\langle \Delta_{\mathbf{k}}^{(s)}\dot{\mathbf{r}}^{(s)}\frac{\partial n_{\mathbf{k}}^{(s)}}{\partial \mathbf{r}} \right\rangle, \tag{9}$$

which allows us to find the expression for $\Lambda^{(s)}$. To extract the Berry curvature contribution to the magnetoconductivity, it is enough to use $\frac{\partial}{\partial \mathbf{k}} = \frac{\partial \epsilon_{\mathbf{k}}}{\partial \mathbf{k}}\frac{\partial}{\partial \epsilon_{\mathbf{k}}} = \mathbf{v}\frac{\partial}{\partial \epsilon_{\mathbf{k}}}$ in the expression for $\Lambda^{(s)}$. We next assume a gradient of temperature, which gives, under an approximation of linear response, a $\frac{\partial n_{\mathbf{k}}^{(s)}}{\partial \mathbf{r}} = -[\nabla T(\mathbf{r})]\frac{\epsilon_{\mathbf{k}}^{(s)}}{T}\frac{\partial f^{(s)}}{\partial \epsilon_{\mathbf{k}}^{(s)}}$ in the expression for the kinetic equation, where $f^{(s)} = \big(e^{\epsilon_{\mathbf{k}}^{(s)}/T} + 1\big)^{-1}$ is an equilibrium Fermi-Dirac distribution function at, in general, nonzero temperature. Expanding in small parameters $\frac{\omega_c}{\mu} < 1$ and $\frac{C}{v} < 1$, we get for $\Lambda^{(s)}$ an expression $\Lambda^{(s)} = \Lambda_{\mathrm{E}}^{(s)} + \Lambda_{\mathrm{T}}^{(s)}$, where

$$\Lambda_{\mathrm{E}}^{(s)} = -s\frac{e^2v^2}{6ck}(\mathbf{EB})\left(\frac{2}{vk}\frac{\partial f}{\partial\epsilon_{\mathbf{k}}} - \frac{\partial^2 f}{\partial\epsilon_{\mathbf{k}}^2}\right) - s\frac{e}{2k}\frac{(\mathbf{EC})}{|C|}\left(1-\frac{v^2}{C^2}+\frac{v\mu}{C^2k}\right) \Theta\left(k-\frac{\mu}{v+|C|}\right)\Theta\left(\frac{\mu}{v-|C|}-k\right), \quad (10)$$

$$\Lambda_{\mathrm{T}}^{(s)} = -s\frac{ev}{6k^2Tc}(\mathbf{B}\nabla T)\left(-2\epsilon_{\mathbf{k}}\frac{\partial f}{\partial\epsilon_{\mathbf{k}}} + vk\frac{\partial f}{\partial\epsilon_{\mathbf{k}}} + vk\epsilon_{\mathbf{k}}\frac{\partial^2 f}{\partial\epsilon_{\mathbf{k}}^2}\right) - \frac{s}{3T}(\mathbf{C}\nabla T)\left(vk\epsilon_{\mathbf{k}}\frac{\partial^2 f}{\partial\epsilon_{\mathbf{k}}^2} + vk\frac{\partial f}{\partial\epsilon_{\mathbf{k}}} + 3\epsilon_{\mathbf{k}}\frac{\partial f}{\partial\epsilon_{\mathbf{k}}}\right), \quad (11)$$

where $\epsilon_{\mathbf{k}} = vk-\mu$ is used. We first observe that in the equations above, $\Lambda^{(+)} = -\Lambda^{(-)}$, and it is consistent with the definition of $\Lambda^{(s)}$ through the collision integral, Eq. (8). In the case when $\tau_{\mathrm{V}} = \infty$, there is no steady state and the distribution function is a function of time. Then, after integrating the kinetic equation over k, we obtain $\frac{\partial N^{(s)}}{\partial t} + \nabla\mathbf{J}^{(s)} = \int[\Lambda_{\mathrm{E}}^{(s)} + \Lambda_{\mathrm{T}}^{(s)}]\frac{k^2dk}{2\pi^2}$, where $N^{(s)} = \int \bar{n}^{(s)}\frac{k^2dk}{2\pi^2}$ and is $\mathbf{J}^{(s)}$ is the local current. Calculations show $\int[\Lambda_{\mathrm{E}}^{(s)} + \Lambda_{\mathrm{T}}^{(s)}]\frac{k^2dk}{2\pi^2} = s\frac{e^2}{4\pi^2c}(\mathbf{EB}) + s\frac{e}{4\pi^2c}g(-\frac{\mu}{T})(\mathbf{B}\nabla T)$, where $g(x) = \frac{xe^x}{e^x+1} - \ln[1+e^x]$. This is the chiral anomaly [8–10], i.e., nonconservation of chiral charge $N^{(+)} - N^{(-)}$ in the presence of electric field or temperature gradient and magnetic field. We note that the tilt alone does not result in the chiral anomaly, but its presence in Eqs. (10) and (11), before momentum integration, is due to the one-dimensional chiral anomaly. When $\tau_{\mathrm{V}} \neq \infty$ and there is a steady state, $\Lambda^{(s)}$ defined in Eq. (8) via the collision integral compensates the chiral anomaly given by Eqs. (10) and (11). Hence, we have the aforementioned $\Lambda^{(s)} = \Lambda_{\mathrm{E}}^{(s)} + \Lambda_{\mathrm{T}}^{(s)}$ equality.

Since $\Lambda^{(s)}$ in steady state is defined by Eqs. (10) and (11), the electron current will obtain magnetic field dependence through the mechanism of the chiral anomaly. To show this, we write an expression for the total electron current with subtracted magnetization parts (see Ref. [6], for example),

$$\mathbf{j} = e\sum_{s=\pm}\int_{\mathbf{k}}\Delta_{\mathbf{k}}^{(s)}\dot{\mathbf{r}}^{(s)}n_{\mathbf{k}}^{(s)} + e\nabla\times T\sum_{s=\pm,\eta=\pm}\int_{\mathbf{k}}\mathbf{\Omega}_{\mathbf{k}\eta}^{(s)}\ln\left[1+e^{-\epsilon_{\mathbf{k}\eta}^{(s)}/T}\right], \quad (12)$$

where for the sake of generality we restored the index η. The second term in the current corresponds to the anomalous Hall effect [4,6,7]. It is very well studied, and we omit it in the following. In order to calculate the magnetoconductivity, we rewrite the kinetic equation as $n_{\mathbf{k}}^{(s)} = \bar{n}^{(s)} + \tau^*[\Lambda^{(s)} - \dot{\mathbf{r}}^{(s)}\frac{\partial n_{\mathbf{k}}^{(s)}}{\partial\mathbf{r}} - \dot{\mathbf{k}}^{(s)}\frac{\partial n_{\mathbf{k}}^{(s)}}{\partial\mathbf{k}}]$, keeping the terms linear in electric field and thermal gradient in the right-hand side only, and expanding them in magnetic field assuming $\frac{\omega_c}{\mu} < 1$ and $\omega_c\tau^* < 1$. The latter condition is used in the Zener-Jones method [34–36] with which one can extract corrections to the distribution function due to the Lorentz force. They are very well studied and we omit them in the following. In order to only extract the chiral anomaly contribution to the current in the limit $\tau_{\mathrm{V}} \gg \tau^*$ valid for large Q momentum splitting of the $s=\pm$ valleys, it is enough to pick $n_{\mathbf{k}}^{(s)} = \bar{n}^{(s)}$. The expression for the magnetic field dependent contribution to the current due to the chiral anomaly is then

$$\delta\mathbf{j}_\Lambda = e\sum_{s=\pm}\int_{\mathbf{k}}\left[\mathbf{v}^{(s)} + \frac{e}{c}(\mathbf{\Omega}_{\mathbf{k}}^{(s)}\mathbf{v}^{(s)})\mathbf{B}\right]\bar{n}^{(s)} = -\frac{2e^2v}{c}\tau_{\mathrm{V}}\int_{\mathbf{k}}\frac{\mathbf{k}}{k}(\mathbf{B}\mathbf{\Omega}_{\mathbf{k}}^{(+)})\Lambda^{(+)} - e\tau_{\mathrm{V}}\mathbf{C}\int_{\mathbf{k}}\Lambda^{(+)}. \quad (13)$$

In the following, we list results for the magnetoconductivity as a sum $\delta\mathbf{j}_\Lambda = \delta\mathbf{j}_{\mathrm{B}}^{[\mathrm{E}]} + \delta\mathbf{j}_{\mathrm{C}}^{[\mathrm{E}]} + \delta\mathbf{j}_{\mathrm{B}}^{[\mathrm{T}]} + \delta\mathbf{j}_{\mathrm{C}}^{[\mathrm{T}]}$.

III. QUADRATIC AND NONANALYTIC MAGNETOCONDUCTIVITY

In this section we are going to study an untilted Dirac system, i.e., $C=0$. To the lowest order in the parameter $\frac{\omega_c}{\mu} < 1$, we get a contribution to the current,

$$\delta\mathbf{j}_{\mathrm{B}}^{[\mathrm{E}]} = \tau_{\mathrm{V}}\frac{e^4v^3}{36\pi^2c^2\mu^2}\left(1+\frac{2\mu^2}{Tv\alpha}e^{-\mu/T}\right)(\mathbf{EB})\mathbf{B}, \quad (14)$$

where $\alpha = \max(\ell^{-1}, L^{-1}, \sqrt{\frac{eB}{c}})$ is the low-k cutoff whose consequences will be discussed below. There, ℓ is the mean free path of an electron, L is the system size, and we assumed $\mu \gg v\alpha$. The obtained magnetoconductivity due to the chiral anomaly qualitatively agrees with the original results of Refs. [11,12]. Importantly, it has a positive sign. Due to the small-k cutoff we observe that when $T > T^* \approx \mu/\ln[\frac{k_F}{\sqrt{eB/c}}]$ given $\sqrt{\frac{eB}{c}}$ is the largest cutoff, the magnetic field dependent correction to the current written in (14) is

$$\delta\mathbf{j}_{\mathrm{B}}^{[\mathrm{E}]} = \tau_{\mathrm{V}}\frac{e^4v^2}{18\pi^2c^2T}e^{-\mu/T}\frac{(\mathbf{EB})\mathbf{B}}{\sqrt{eB/c}}, \quad (15)$$

which is now proportional to the $\frac{3}{2}$ power of the magnetic field, i.e., it is a nonanalytic correction to the current. This contribution to the current is not from the Fermi surface but rather from the $k \sim 0$ region, where the Berry curvature is singular. As we show, the regime of this result is magnetic field and impurity concentration dependent. Namely, in Weyl semimetals, $\ell^{-1} = N\mu^2$, where N is a parameter characterizing impurity concentration. For the existence of the regime we then write a condition $N\mu^2 < \ell_{\mathrm{B}}^{-1} \ll k_{\mathrm{F}}$. The latter inequality is satisfied by both the Ioffe-Regel condition of a good metal, $k_{\mathrm{F}}\ell > 1$, and by the assumption of small magnetic fields $\frac{\omega_c}{\mu} < 1$. The former inequality can be achieved by either decreasing the parameter N or increasing the magnetic field, while making sure the $\frac{\omega_c}{\mu} < 1$ condition is always satisfied.

The magnetic field dependence of the current driven by the temperature gradient (thermoelectric effect) due to the chiral anomaly is

$$\delta\mathbf{j}_{\mathrm{B}}^{[\mathrm{T}]} = \tau_{\mathrm{V}}\frac{e^3v^3}{36\pi^2c^2\mu}\left[\frac{2\pi^2}{3}\left(\frac{T}{\mu}\right)^2 + \frac{2\mu^2}{T(v\alpha)}e^{-\mu/T}\right]\frac{(\mathbf{B}\nabla T)\mathbf{B}}{T}. \quad (16)$$

Again, when $\sqrt{\frac{eB}{c}}$ is the largest cutoff, and when the temperature is $T > T^* \approx \mu/\ln[\frac{k_F}{\sqrt{eB/c}}]$, we get nonanalytic magnetic field dependence of the current,

$$\delta \mathbf{j}_{\mathrm{B}}^{[\mathrm{T}]} = \tau_{\mathrm{V}} \frac{e^3 v^2 \mu}{18\pi^2 c^2 T} e^{-\mu/T} \frac{(\mathbf{B}\nabla T)\mathbf{B}}{T\sqrt{eB/c}}, \tag{17}$$

consistent with the obtained results for the magnetoconductivity.

Formally, the divergence of integrals in the expressions for the currents (14) and (16) signals the inapplicability of the semiclassical equations of motion (4) at small k, hence the need for a cutoff [see the discussion after Eq. (4)].

IV. LINEAR MAGNETOCONDUCTIVITY

In all previous derivations, the terms linear in magnetic field dropped out due to angle integration. Let us now add a tilt to the spectrum, $C_s \neq 0$, as defined in the Hamiltonian (1). The tilt and magnetic field dependent contribution to the current driven by the electric field is

$$\delta \mathbf{j}_{\mathrm{C}}^{[\mathrm{E}]} = \tau_{\mathrm{V}} \frac{e^3}{\pi^2 c}\left[\frac{1}{9}(\mathbf{E}\mathbf{C})\mathbf{B} + \frac{1}{4}(\mathbf{E}\mathbf{B})\mathbf{C}\right], \tag{18}$$

whose direction is in the direction of the magnetic field, and along the direction of the tilt. We generalized the direction of the tilt by a $C\mathbf{e}_z \to \mathbf{C}$ replacement. The electric current driven by the temperature gradient has similar corrections, namely,

$$\delta \mathbf{j}_{\mathrm{C}}^{[\mathrm{T}]} = -\tau_{\mathrm{V}} \frac{e^2}{\pi^2 c} g\left(-\frac{\mu}{T}\right)\left[\frac{1}{9}(\mathbf{C}\nabla T)\mathbf{B} + \frac{1}{4}(\mathbf{B}\nabla T)\mathbf{C}\right], \tag{19}$$

where $g(x)$ is defined after Eq. [11], and $g(-x) \approx -xe^{-x}$ for $x \gg 1$.

The tilt, as introduced in expression (1), breaks time-reversal symmetry, therefore the linear magnetoconductivity is not prohibited by the Onsager relations for the conductivity. We note that in deriving the results (18) and (19), we approximated the scattering time to be isotropic, which is a valid assumption for $v \gg |C|$. We note that only in the case of an antisymmetric tilt $C_s = sC$ can we approximate the kinetic equation in an analytic way, and only in this case the $\Lambda^{(+)} = -\Lambda^{(-)}$ relation holds. This relation is consistent with the steady state approximation of the kinetic equation. If the tilt is such that $C_+ \neq -C_-$, there will be no steady state in the system and the distribution function will be a function of time. Finally, we note that the first terms in (18) and (19) are due to the last two terms of expressions (10) and (11), correspondingly. We called them the terms which resulted from the one-dimensional chiral anomaly [10].

V. CONCLUSIONS

We have studied various magnetotransport properties of a Weyl semimetal system due to the chiral anomaly. Apart from the predicted quadratic magnetoconductivity in Refs. [11–13], we find a $\propto B^{3/2}$ nonanalytic contribution to the magnetoconductivity at finite temperatures [see Eqs. (14) and (15)]. This is due to the singular nature of the Berry curvature in Weyl semimetals.

We have shown that the tilt of the Dirac cones results in two contributions to the magnetoconductivity that are linear in magnetic field. The first contribution has a direction in the magnetic field, and the second one in the direction of the tilt [see Eqs. (18)]. The tilt of the Dirac cones is responsible for a mechanism that is similar to the one-dimensional chiral anomaly, however, its contribution to the three-dimensional chiral anomaly vanishes. The tilt breaks time-reversal symmetry, therefore, linear magnetoconductivity is not prohibited.

We have studied the magnetic field dependence of the current driven by the temperature gradient (thermoelectric effect) in a Weyl semimetal due to the chiral anomaly. The results are qualitatively consistent with the ones obtained for magnetoconductivity [see Eqs. (16), (17), and (19)].

In the present work we have developed an analytical treatment of the electron-impurity collision integral for a Weyl semimetal under the assumption of short range impurities. We have shown that intervalley scattering processes cannot be ignored and are responsible for the magnetotransport in Weyl semimetals due to the chiral anomaly. A number of papers, for example, Refs. [37–40], ignore these processes by setting $\tau_{\mathrm{V}} = \infty$ in Eq. (7), while Refs. [11,12,41] do include these processes. The approach of the present paper qualitatively agrees only with Ref. [12]. Magnetic field dependent contributions to the currents obtained in Refs. [37–40] come from the $\propto\tau^*$ corrections to the distribution function [see the discussion after Eq. (12)]. These corrections were ignored in the present paper because of their parametric smallness due to the $\tau_{\mathrm{V}} \gg \tau^*$ assumption.

We note that contributions linear in magnetic field to the current of untilted Weyl semimetals were obtained in Ref. [40]. Their origin is in the interplay of orbital magnetization, Lorentz force, and quadratic corrections to the Dirac spectrum. References [38,39] also studied the magnetothermal properties of Weyl semimetals. The low-k divergence of the integrals defining the thermoelectric coefficients were ignored in them.

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